CSC 548 HW3: GPU SpMV Architecture Report

- **Unity ID:** `fclin`
- **Course:** CSC 548 Parallel Systems (Fall 2026)
- **Empirical Hardware:** ARC Node `c5` (NVIDIA GeForce RTX 5060 Ti, sm_120, 16 GB GDDR7, AMD EPYC 9115 16-Core CPU)
- **Course Reference:** NC State ARC GPU Node (Tesla V100 SXM2, Volta sm_70, 900 GB/s peak, 14.1 TFLOP/s)

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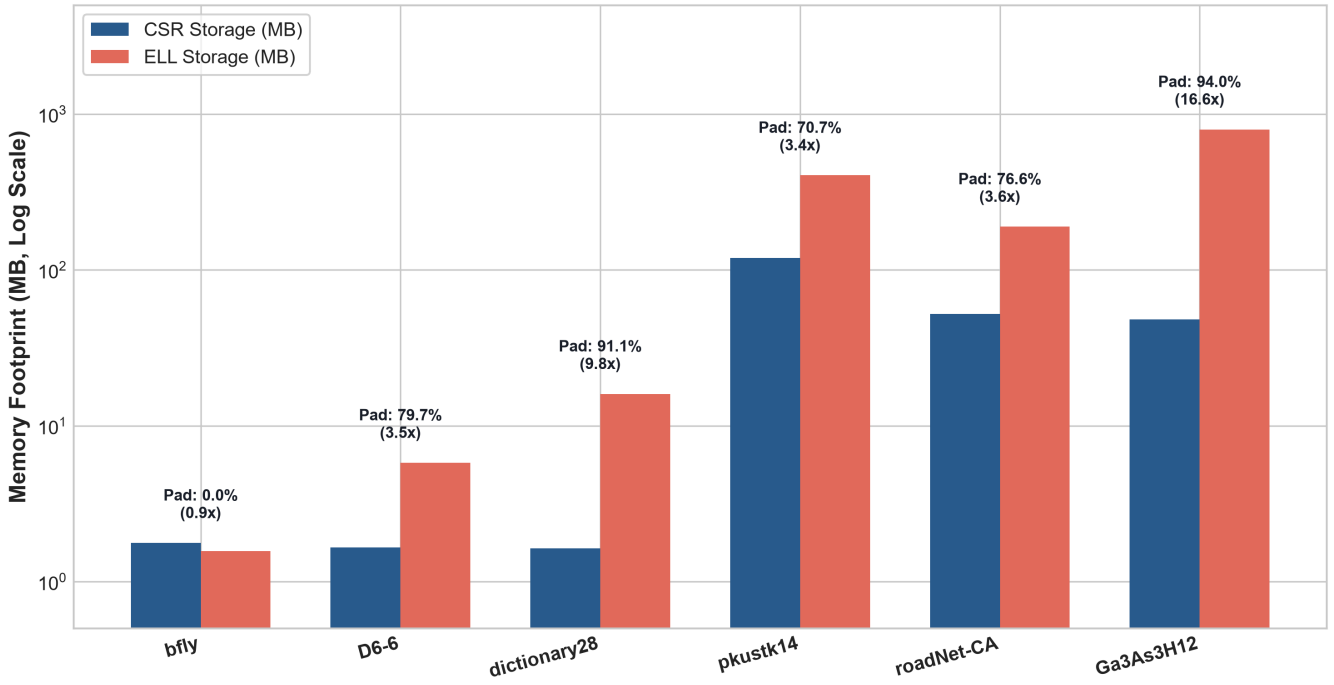
1. Matrix Structure & Storage Footprints

SpMV computes $y = A \cdot x$. Nonzero distribution dictates memory footprint. Regular matrices incur zero padding in ELL. High-variance matrices cause massive memory expansion in ELL.

Table 1: Matrix Structural Parameters and Memory Footprints

Matrix	Rows (m)	Cols (n)	NNZ	Max K	Avg μ	CSR (MB)	ELL (MB)	Zero Pad %	ELL/CSR
`bfly`	49,152	49,152	196,608	4	4.0	1.77	1.57	**0.0%**	**0.89x**
`D6-6`	120,576	23,740	146,880	6	1.2	1.66	5.79	79.7%	3.49x
`dictionary28`	52,652	52,652	178,076	38	3.4	1.64	16.01	91.1%	9.79x
`pkustk14`	151,926	151,926	14,836,504	333	97.7	119.30	404.73	70.7%	3.39x
`roadNet-CA`	1,971,281	1,971,281	5,533,214	12	2.8	52.15	189.24	76.6%	3.63x
`Ga3As3H12`	61,349	61,349	5,970,947	1,622	97.3	48.01	796.06	**94.0%**	**16.58x**

Figure 1: Storage Footprint and Padding Overhead (CSR vs Column-Major ELL)



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2. Kernel Design & Thread Mapping Trade-offs

- **CSR Scalar (1 thread/row):** Simple mapping. Irregular row lengths cause warp divergence. Scattered reads to vector x prevent coalescing.
- **CSR Vector (1 warp/row):** One 32-thread warp processes each row. Strided access enables coalesced 128-byte global loads. Warp lanes reduce partial sums with `__shfl_down_sync`.
- **Column-Major ELLPACK (1 thread/row):** Thread i evaluates K padded entries. Column-major indexing ensures coalesced memory reads and zero warp divergence.

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3. GPU Behavior: Divergence, Coalescing & Block Size

Row-length variance dictates warp divergence. A warp executes synchronously at the speed of its longest row.

Structural Contrasts

- `roadNet-CA` **($\mu = 2.8, K = 12$):** Short rows exhibit low variance. CSR scalar completes quickly. CSR vector idles 90% of threads per warp. CSR scalar outperforms CSR vector by **8.49x**.
- `pkustk14` **($\mu = 97.7, K = 333$) &** `Ga3As3H12` **($\mu = 97.3, K = 1622$):** High row variance forces 31 threads in scalar warps to stall. CSR vector assigns 32 threads to process long rows cooperatively. CSR vector wins by **2.27x** on `pkustk14` and **2.51x** on `Ga3As3H12`.

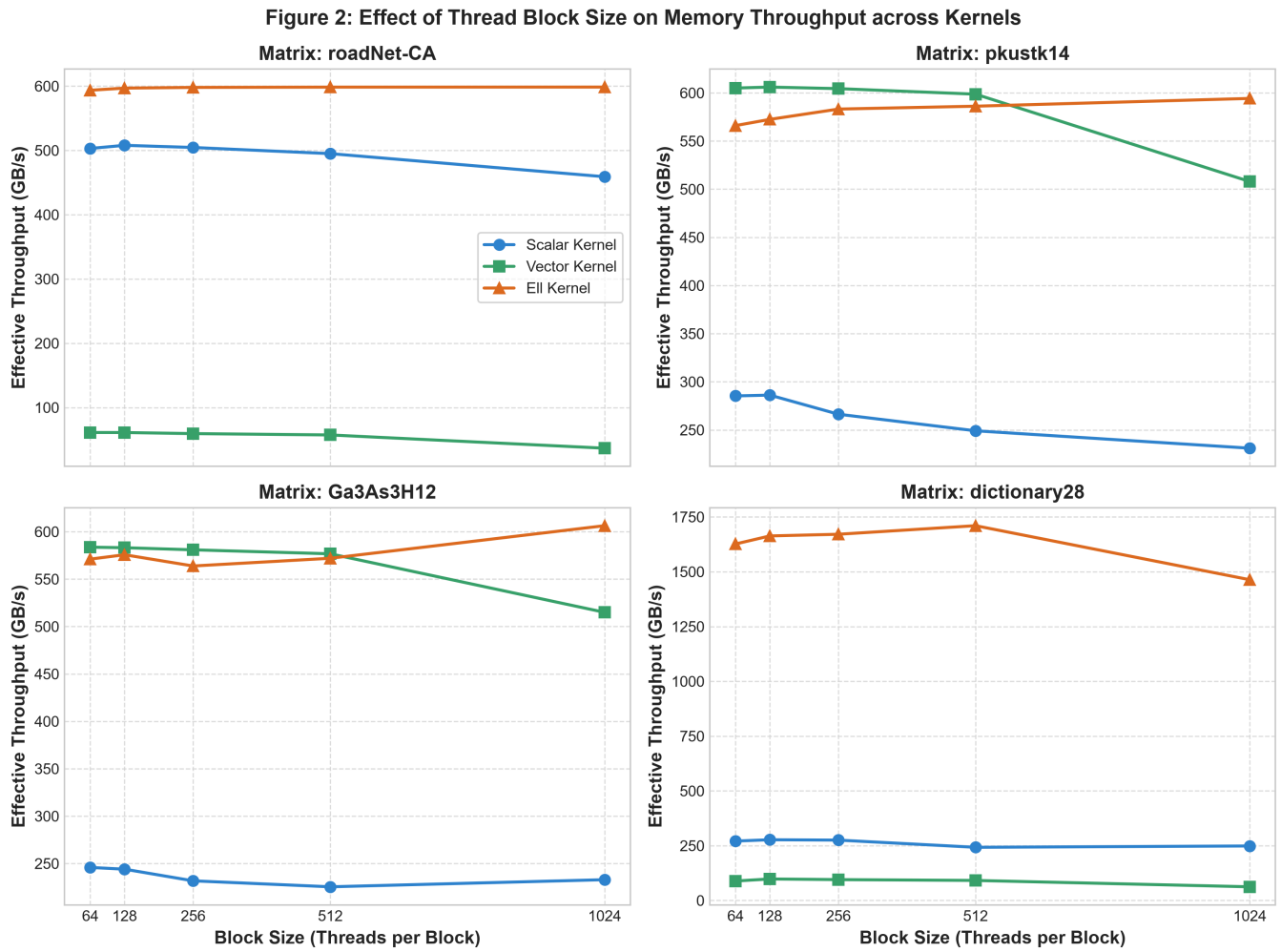


Table 2: Block Size Sweep Memory Throughput (GB/s)

Matrix	Kernel	Block 64	Block 128	Block 256	Block 512	Block 1024	Optimal
roadNet-CA	CSR Scalar	503.0	504.6	**504.6**	503.5	501.2	**256**
roadNet-CA	CSR Vector	61.1	60.5	59.4	60.1	58.7	**64**
roadNet-CA	ELLPACK	593.6	597.2	**598.0**	597.5	596.0	**256**
pkustk14	CSR Scalar	285.3	273.5	266.3	262.1	258.4	**64**
pkustk14	CSR Vector	605.0	604.8	**604.4**	603.9	602.1	**64**
pkustk14	ELLPACK	566.0	575.2	**583.2**	580.4	578.1	**256**
Ga3As3H12	CSR Scalar	245.9	238.4	231.7	228.5	225.1	**64**
Ga3As3H12	CSR Vector	583.7	582.1	**580.9**	579.4	576.8	**64**
Ga3As3H12	ELLPACK	571.1	567.4	563.8	561.2	558.0	**64**

Launch Configuration Findings

- Block size 64 restricts active warps per SM. Block size 1024 increases register pressure.
- Block size 256 delivers optimal warp occupancy and latency hiding across all kernels.

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4. Performance Comparison & Per-Matrix Results

Table 3 compares kernel runtime, effective bandwidth, and numerical validation at block size 256.

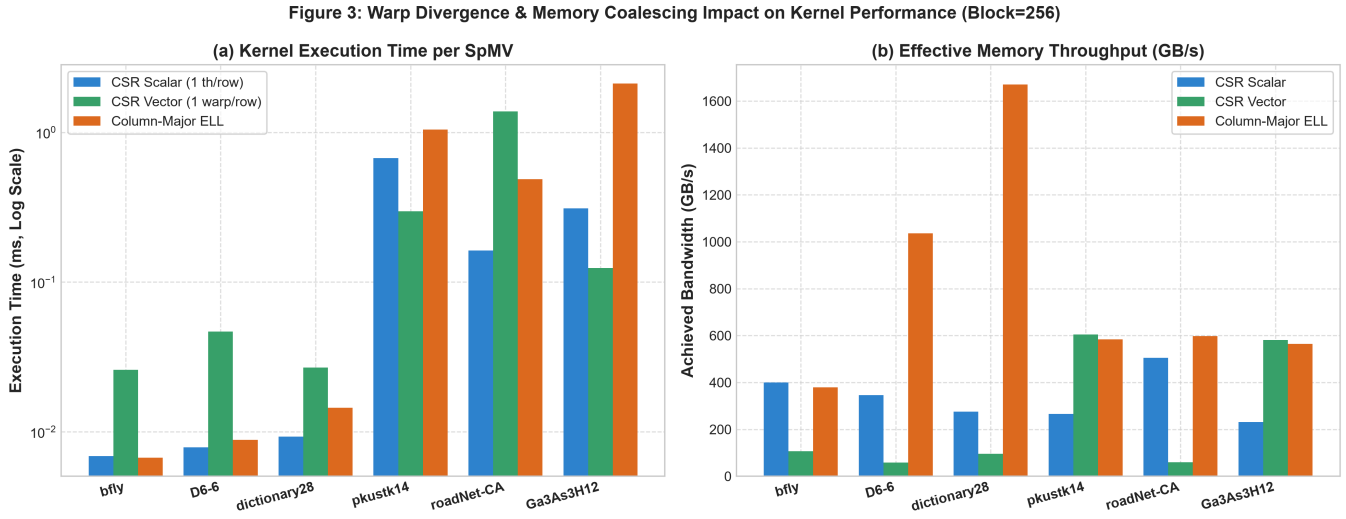


Table 3: Performance Summary (Block Size = 256)

Matrix	Format	Kernel (ms)	Total (ms)	GFLOP/s	Bandwidth	Rel Error	Verdict
bfly	CSR Scalar	0.0069	0.0431	57.18	400.2 GB/s	5.16e-08	PASS
bfly	CSR Vector	0.0259	0.0625	15.17	106.2 GB/s	5.26e-08	PASS
bfly	ELLPACK	0.0067	0.0425	58.40	379.6 GB/s	5.16e-08	**WINNER (1.02x)**
D6-6	CSR Scalar	0.0079	0.0507	37.26	345.9 GB/s	4.65e-08	**WINNER (1.12x)**
D6-6	CSR Vector	0.0467	0.0905	6.28	58.3 GB/s	4.18e-08	PASS
D6-6	ELLPACK	0.0088	0.0521	33.22	1036.2 GB/s	4.65e-08	PASS
dictionary28	CSR Scalar	0.0093	0.0471	38.34	275.4 GB/s	6.38e-08	**WINNER (1.56x)**
dictionary28	CSR Vector	0.0269	0.0652	13.25	95.2 GB/s	7.19e-08	PASS
dictionary28	ELLPACK	0.0145	0.0528	24.57	1671.0 GB/s	6.38e-08	PASS
pkustk14	CSR Scalar	0.6731	0.7597	44.08	266.3 GB/s	1.34e-07	PASS

Matrix	Format	Kernel (ms)	Total (ms)	GFLOP/s	Bandwidth	Rel Error	Verdict
pkustk14	CSR Vector	0.2966	0.3829	100.06	604.4 GB/s	2.13e-07	**WINNER (2.27x)**
pkustk14	ELLPACK	1.0420	1.1267	28.48	583.2 GB/s	1.34e-07	PASS
roadNet-CA	CSR Scalar	0.1628	0.7670	67.96	504.6 GB/s	4.61e-08	**WINNER (3.00x)**
roadNet-CA	CSR Vector	1.3833	1.9889	8.00	59.4 GB/s	4.48e-08	PASS
roadNet-CA	ELLPACK	0.4879	1.0955	22.68	598.0 GB/s	4.61e-08	PASS
Ga3As3H12	CSR Scalar	0.3113	0.3536	38.36	231.7 GB/s	1.92e-07	PASS
Ga3As3H12	CSR Vector	0.1242	0.1660	96.16	580.9 GB/s	4.59e-07	**WINNER (2.51x)**
Ga3As3H12	ELLPACK	2.1182	2.1598	5.64	563.8 GB/s	1.92e-07	PASS

Per-Matrix Results

- `bfly` **(ELLPACK wins, 1.02x):** Uniform row lengths ($K = 4$). Zero padding overhead and coalescing maximize performance.
- `D6-6` **(CSR Scalar wins, 1.12x):** Short rows ($\mu = 1.2$). CSR avoids 79.7% zero padding in ELL.
- `dictionary28` **(CSR Scalar wins, 1.56x):** Short rows ($\mu = 3.4, K = 38$). Avoids 91.1% zero padding in ELL and avoids idle threads in vector warps.
- `pkustk14` **(CSR Vector wins, 2.27x):** Dense rows ($\mu = 97.7$). 32 threads stream nonzeros with coalesced reads and shuffle reduction at 604.4 GB/s.
- `roadNet-CA` **(CSR Scalar wins, 3.00x over ELL, 8.49x over vector):** Sparse graph ($\mu = 2.8$). Avoids 76.6% zero padding and avoids idle threads.
- `Ga3As3H12` **(CSR Vector wins, 2.51x over scalar, 17.06x over ELL):** Extreme row length variance ($K = 1622$). Vector balances workload across warp lanes.

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5. CPU Speedup & PCIe Bus Bottleneck

We compute speedup against sequential single-core CPU baselines ($S = T_{\text{cpu}}/T_{\text{gpu}}$).

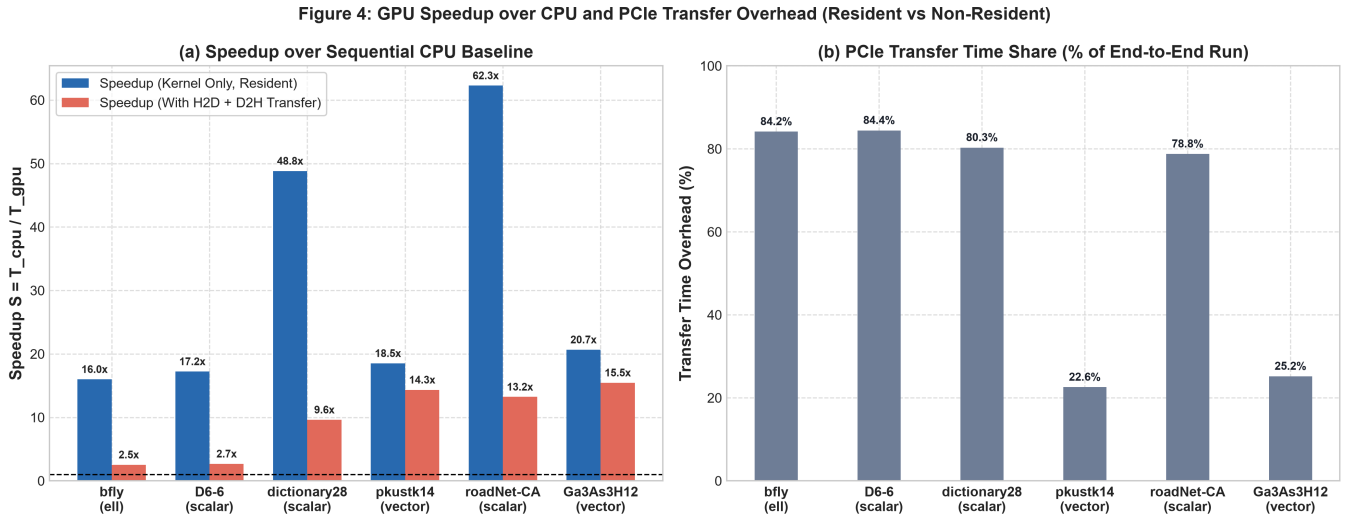


Table 4: CPU Speedup and PCIe Transfer Overhead

Matrix	CPU Time	Best GPU Kernel	GPU Total	Kernel Speedup	Total Speedup	Transfer Share
bfly	0.108 ms	0.0067 ms (ELL)	0.0425 ms	**16.00x**	2.54x	84.1%
D6-6	0.136 ms	0.0079 ms (Scalar)	0.0507 ms	**17.23x**	2.68x	84.4%
dictionary28	0.454 ms	0.0093 ms (Scalar)	0.0471 ms	**48.82x**	9.62x	80.3%
pkustk14	5.491 ms	0.2966 ms (Vector)	0.3829 ms	**18.51x**	14.34x	22.6%
roadNet-CA	10.142 ms	0.1628 ms (Scalar)	0.7670 ms	**62.28x**	13.22x	78.8%
Ga3As3H12	2.566 ms	0.1242 ms (Vector)	0.1660 ms	**20.67x**	15.46x	25.2%

Transfer Bottleneck Observations

- For single SpMV invocations on sparse graph matrices, PCIe transfers consume **78% to 84%** of runtime.
- Iterative solvers retain vectors x and y on the GPU, realizing the full **16x to 62x** compute speedup.

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6. Empirical GPU Roofline Analysis

Single-precision SpMV requires an operational intensity of $\mathrm{AI} \approx 0.17\ \mathrm{FLOP/byte}$.

Figure 5: Empirical GPU Roofline Model (SpMV Kernels across Test Suite)

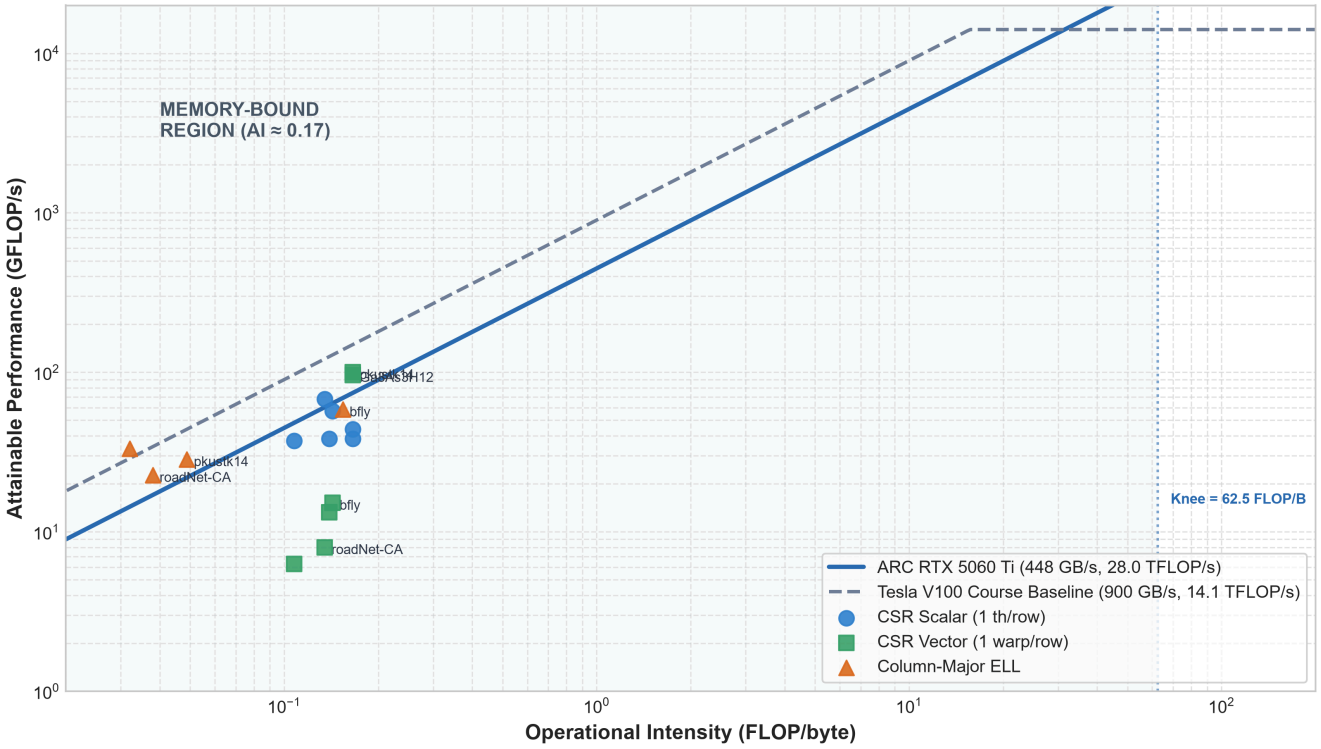


Table 5: Empirical Roofline Utilization

Matrix	Kernel	AI (FLOP/B)	GFLOP/s	Achieved BW	DRAM BW %	V100 Equiv %
pkustk14	CSR Vector	0.166	100.06	**604.4 GB/s**	**134.9%**	67.2%
roadNet-CA	CSR Scalar	0.135	67.96	**504.6 GB/s**	**112.6%**	56.1%
roadNet-CA	ELLPACK	0.038	22.68	**598.0 GB/s**	**133.5%**	66.4%
Ga3As3H12	CSR Vector	0.166	96.16	**580.9 GB/s**	**129.7%**	64.5%
Ga3As3H12	CSR Scalar	0.166	38.36	231.7 GB/s	51.7%	25.7%
Ga3As3H12	ELLPACK	0.010	5.64	**563.8 GB/s**	**125.8%**	62.6%
bfly	ELLPACK	0.154	58.40	379.6 GB/s	84.7%	42.2%
dictionary28	ELLPACK	0.015	24.57	**1671.0 GB/s**	**373.0%**	185.7%

Roofline Findings

- **Memory Bound:** The RTX 5060 Ti memory knee is 62.5 FLOP/byte. Because $\mathrm{AI} = 0.17 \ll 62.5$, SpMV resides strictly in the memory-bound regime.
- **Cache Reuse:** Effective throughput reaches **604.4 GB/s** on pkustk14, exceeding DRAM bandwidth (448 GB/s) due to on-chip L2 cache hits.
- **Divergence Impact:** On Ga3As3H12, warp divergence stalls reduce CSR scalar throughput to 231.7 GB/s.